

NILAKSHI MISHRA

AI/ML Engineer | RAG | LLM | Backend Developer
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SUMMARY

AI/ML & Backend Developer with 1.5+ years of professional experience building AI-powered applications using LLMs, RAG, AI Agents, and event-driven architectures. Skilled in Python, FastAPI, Redis, MongoDB, PostgreSQL, REST APIs, and AWS, with hands-on experience developing AI chatbots, automated workflows, and scalable backend systems.

PROFESSIONAL EXPERIENCE

SOFTWARE AI/ML ENGINEER | Ommify Technologies Private Limited

Jan2025 – Present

- Built LLM-based chatbot with tool calling to automate marketing campaigns via single prompt.
- Developed event-driven notification system using Redis queue and LLM-based classification.
- Designed MongoDB → Redis pipeline using producer-consumer architecture.
- Deployed containerized applications on AWS ECS (EC2) with VPC and ALB.
- Implemented secure networking (private subnets, NAT Gateway, Security Groups).
- Integrated PostgreSQL (RDS), Redis, and APIs for scalable backend systems.

KEY PROJECTS-

AI Campaign Chatbot

Python, FastAPI, OpenAI GPT, RAG, FAISS

- Built RAG-based chatbot with tool calling for campaign automation.

Event-Driven Notification Service

Python, FastAPI, PostgreSQL, Redis, Scheduler, Email & WhatsApp API

- Built Redis-based event processing and automated seller notifications.

Enterprise Scraper Bot Factory

Python, Celery, Redis, OpenSearch, MinIO

- Built distributed scraping pipelines with asynchronous processing and data indexing.

SEO Optimization Tool

Python, FastAPI, PostgreSQL, AI/LLM

- Developed backend APIs for SEO analysis and AI-powered content recommendations.

AI Intern | CETPA Infotech Pvt. Ltd.

Sep2024 – Jan2025

- Gained hands-on experience in Python, Data Analysis, and Machine & Deep Learning.
- Worked on data preprocessing, model training, and evaluation & Built projects using Pandas, NumPy & Matplotlib.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, C
- **Frameworks:** React.js, HTML, CSS, FastAPI
- **AI/ML :** LLM, RAG, LangChain, AI Agents, Tool Calling, MCP
- **DA Libraries:** NumPy, Pandas, Matplotlib.
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, minIO
- **Vector Databases:** Pinecone, FAISS
- **Cloud & DevOps:** AWS (EC2, ECS, VPC, ALB), Docker
- **Tools & Platforms:** Git, GitHub, Bitbucket, VS Code, Jupyter.

RESEARCH AND PUBLICATIONS

- Published research on a triple-band miniaturized antenna (HFSS) for Bluetooth, WiMAX, and mobile communication.
- Presented at ICVMWT 2024; demonstrated performance via radiation pattern and surface current analysis (Springer).
- Achieved efficient performance via radiation pattern and surface current analysis.

EDUCATION

Bachelor of Technology | Electronics and Communication Engineering

2020-2024

Dr. Ram Manohar Lohiya Awadh University, Ayodhya-75%